Geon Lee

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RESEARCH SUMMARY

My research explores how interactions among relational data, retrieved knowledge, and AI agents shape learning, reasoning, and behavior. By uncovering their underlying principles, I aim to develop intelligent systems that effectively leverage diverse information while remaining reliable, generalizable, and robust in real-world settings.

WORK EXPERIENCE

KAIST — Seoul, South Korea
Postdoctoral Research Fellow (Advisor: Prof. Kijung Shin) — Mar. 2026 – Present

Snap Research — Bellevue, WA, USA
Research Intern (Mentors: Liam Collins, Clark Ju, Bhuvesh Kumar, Tong Zhao | Manager: Neil Shah) — Apr. – Jun. 2025
Data Augmentation for Generative Recommendation

NEC Labs America — Princeton, NJ, USA
Research Intern (Mentor: Wenchao Yu | Manager: Haifeng Chen) — May – Aug. 2023
Multi-Modal Time Series Forecasting

Amazon — San Francisco, CA, USA
Applied Scientist Intern (Mentor: Zhonghao Luo | Manager: Tao Ye) — Sep. – Dec. 2022
Conversational Music Recommendation for Alexa AI

EDUCATION

KAIST — Seoul, South Korea
M.S. & Ph.D. in Artificial Intelligence — Sep. 2020 – Feb. 2026
Advisor: Prof. Kijung Shin
PhD Dissertation Award, KAIST College of Engineering

Sungkyunkwan University — Suwon, South Korea
B.S. in Computer Science and Engineering — Mar. 2016 – Aug. 2019
GPA: 4.41/4.50; Major GPA: 4.45/4.50
Ranked 1st in the College of CSE

TUTORIALS

[1] Retrieval-Augmented Generation for LLM-Based Recommender Systems
Sunwoo Kim*, Geon Lee*, Kyungho Kim*, Liam Collins, Neil Shah, and Kijung Shin
CIKM 2026 & **KDD 2026**

[2] Mining of Real-World Hypergraphs: Patterns, Tools, and Generators
Geon Lee, Jaemin Yoo, and Kijung Shin
KDD 2023, **WWW 2023**, **ICDM 2022**, & **CIKM 2022**

PUBLICATIONS

[1] Zero-Shot Tabular Learning via Hypergraph Reasoning with LLMs
Geon Lee, Kyungho Kim, Dongwon Choi, Juyeon Kim, Shinhwan Kang, and Kijung Shin
EMNLP 2026

[2] Training-Free LLM-Based Recommendation with Post-LLM Item Refinement Using Collaborative Signals
Kyungho Kim, Sunwoo Kim, Geon Lee, Shinhwan Kang, Sojeong Kim, Liam Collins, Bhuvesh Kumar, Donald Loveland, and Kijung Shin
CIKM 2026 (Short Paper)

[3] Toward a Science of AI Agent Societies
Geon Lee*, Fanchen Bu*, Soo Yong Lee*, Sunwoo Kim, and Kijung Shin
KDD 2026 (Blue Sky)

[4] Rethinking Contrastive Learning for Graph Collaborative Filtering: Limitations and A Simple Remedy
Geon Lee, Sunwoo Kim, Kyungho Kim, and Kijung Shin
ICML 2026

[5] Hybrid-Vector Retrieval for Visually Rich Documents: Combining Single-Vector Efficiency and Multi-Vector Accuracy
Juyeon Kim, Geon Lee, Dongwon Choi, Taeuk Kim, and Kijung Shin
ACL 2026 (Findings)

[6] ItemRAG: Item-Based Retrieval-Augmented Generation for LLM-Based Recommendation
Sunwoo Kim, Geon Lee, Kyungho Kim, Jaemin Yoo, and Kijung Shin
SIGIR 2026 (Short Paper)

[7] ReFuGe: Feature Generation for Prediction Tasks on Relational Databases with LLM Agents
Kyungho Kim, Geon Lee, Juyeon Kim, Dongwon Choi, Shinhwan Kang, and Kijung Shin
WWW 2026 (Short Paper)

[8] Sequential Data Augmentation for Generative Recommendation
Geon Lee, Bhuvesh Kumar, Clark Mingxuan Ju, Tong Zhao, Kijung Shin, Neil Shah, and Liam Collins
WSDM 2026

[9] Revisiting LightGCN: Unexpected Inflexibility, Inconsistency, and A Remedy Towards Improved Recommendation
Geon Lee, Kyungho Kim, Fanchen Bu, Langzhang Liang, and Kijung Shin
Transactions on Recommender Systems (2026) *Invited Extended Version of the RecSys 2024 Paper*

[10] From Binary to General Attributes: Attributed Hypergraph Generation with Realistic Interplay Between Structure and Attributes
Jaewan Chun, Seokbum Yoon, Minyoung Choe, Geon Lee, and Kijung Shin
Knowledge and Information Systems (SCIE, 2026) *Invited Extended Version of the ICDM 2025 Paper*

[11] A Comprehensive Survey on Multi-Behavior Recommender Systems: Extended Taxonomy and Recent Advances
Kyungho Kim, Sunwoo Kim, Geon Lee, Jinhong Jung, and Kijung Shin
International Journal of Data Science and Analytics (2026) *Invited to the PAKDD 2025 Special Issue*

[12] RDB2G-Bench: A Comprehensive Benchmark for Automatic Graph Modeling of Relational Databases
Dongwon Choi, Sunwoo Kim, Juyeon Kim, Kyungho Kim, Geon Lee, Shinhwan Kang, Myunghwan Kim, and Kijung Shin
NeurIPS 2025 (Dataset & Benchmark)

[13] TimeXL: Explainable Multi-modal Time Series Prediction with LLM-in-the-Loop
Yushan Jiang, Wenchao Yu, Geon Lee, Dongjin Song, Kijung Shin, Wei Cheng, Yanchi Liu, and Haifeng Chen
NeurIPS 2025

[14] Attributed Hypergraph Generation with Realistic Interplay Between Structure and Attributes
Jaewan Chun, Seokbum Yoon, Minyoung Choe, Geon Lee, and Kijung Shin
ICDM 2025 *Best Paper Award; Best-Ranked Paper*

[15] Identifying Group Anchors in Real-World Group Interactions Under Label Scarcity
Fanchen Bu, Geon Lee, Minyoung Choe, and Kijung Shin
ICDM 2025

[16] A Self-Supervised Mixture-of-Experts Framework for Multi-behavior Recommendation
Kyungho Kim, Sunwoo Kim, Geon Lee, and Kijung Shin
CIKM 2025

[17] SkySearch: Satellite Video Search at Scale
Minyoung Choe*, Geon Lee*, Changhun Han*, Suji Kim, Woong Hu, Hyebeen Hwang, Geunseok Park, Byeongyeon Kim, Hyesook Lee, Ha-Myung Park, and Kijung Shin
KDD 2025 (Applied Data Science)

[18] KGMEL: Knowledge Graph-Enhanced Multimodal Entity Linking
Juyeon Kim, Geon Lee, Taeuk Kim, and Kijung Shin
SIGIR 2025 (Short Paper)

[19] MARIOH: Multiplicity-Aware Hypergraph Reconstruction
Kyuhan Lee, Geon Lee, and Kijung Shin
ICDE 2025

[20] Multi-Behavior Recommender Systems: A Survey
Kyungho Kim, Sunwoo Kim, Geon Lee, Jinhong Jung, and Kijung Shin
PAKDD 2025 (Survey) *Best Survey Paper Award*

[21] Beyond Neighbors: Distance-Generalized Graphlets for Enhanced Graph Characterization
Yeongho Kim, Yuyeong Kim, Geon Lee, and Kijung Shin
WWW 2025

[22] TimeCAP: Learning to Contextualize, Augment, and Predict Time Series Events with Large Language Model Agents
Geon Lee, Wenchao Yu, Kijung Shin, Wei Cheng, and Haifeng Chen
AAAI 2025

[23] A Survey on Hypergraph Mining: Patterns, Tools, and Generators
Geon Lee*, Fanchen Bu*, Tina Eliassi-Rad, and Kijung Shin
ACM Computing Surveys (SCI, 2025)

[24] Resource2Box: Learning to Rank Resources in Distributed Search Using Box Embedding
Ulugbek Ergashev, Geon Lee, Kijung Shin, Eduard Dragut, and Weiyi Meng
ICDM 2024

[25] Revisiting LightGCN: Unexpected Inflexibility, Inconsistency, and A Remedy Towards Improved Recommendation
Geon Lee, Kyungho Kim, and Kijung Shin
RecSys 2024 (Short Paper) *Best Short Paper Candidate*

[26] Post-Training Embedding Enhancement for Long-Tail Recommendation
Geon Lee, Kyungho Kim, and Kijung Shin
CIKM 2024 (Short Paper)

[27] Towards Better Utilization of Multiple Views for Bundle Recommendation
Kyungho Kim, Sunwoo Kim, Geon Lee, and Kijung Shin
CIKM 2024 (Short Paper)

[28] Representative and Back-in-Time Sampling from Real-World Hypergraphs
Minyoung Choe, Jaemin Yoo, Geon Lee, Woonsung Baek, U Kang, and Kijung Shin
Transactions on Knowledge Discovery from Data (SCIE, 2024) *Invited Extended Version of the WWW 2022 Paper*

[29] VilLain: Self-Supervised Learning on Homogeneous Hypergraphs without Features via Virtual Label Propagation
Geon Lee, Soo Yong Lee, and Kijung Shin
WWW 2024

[30] Random Walk with Restart on Hypergraphs: Fast Computation and an Application to Anomaly Detection
Jaewan Chun, Geon Lee, Kijung Shin, and Jinhong Jung
Data Mining and Knowledge Discovery (SCIE, 2023)

[31] Hypergraph Motifs and Their Extensions Beyond Binary
Geon Lee*, Seokbum Yoon*, Jihoon Ko, Hyunju Kim, and Kijung Shin
The VLDB Journal (SCIE, 2023) *KAIST Outstanding Paper Award*

[32] Hypercore Decomposition for Non-Fragile Hyperedges: Concepts, Algorithms, Observations, and Applications
Fanchen Bu, Geon Lee, and Kijung Shin
Data Mining and Knowledge Discovery (SCIE, 2023)

[33] Temporal Hypergraph Motifs
Geon Lee and Kijung Shin
Knowledge and Information Systems (SCIE, 2023) *Invited Extended Version of the ICDM 2022 Paper*

[34] Set2Box: Similarity Preserving Representation Learning for Sets
Geon Lee, Chanyoung Park, and Kijung Shin
ICDM 2022

[35] HashNWalk: Hash and Random Walk Based Anomaly Detection in Hyperedge Streams
Geon Lee, Minyoung Choe, and Kijung Shin
IJCAI 2022

[36] MiDaS: Representative Sampling from Real-World Hypergraphs
Minyoung Choe, Jaemin Yoo, Geon Lee, Woonsung Baek, U Kang, and Kijung Shin
WWW 2022

[37] Simple Epidemic Models with Segmentation Can Be Better than Complex Ones
Geon Lee, Se-eun Yoon, and Kijung Shin
PLOS ONE (SCIE, 2022) *Oral, epiDAMIK @ KDD 2021*

[38] THyMe+: Temporal Hypergraph Motifs and Fast Algorithms for Exact Counting
Geon Lee and Kijung Shin
ICDM 2021 *Best-Ranked Paper*

[39] How Do Hyperedges Overlap in Real-World Hypergraphs? — Patterns, Measures, and Generators
Geon Lee*, Minyoung Choe*, and Kijung Shin
WWW 2021

[40] Hypergraph Motifs: Concepts, Algorithms, and Discoveries
Geon Lee, Jihoon Ko, and Kijung Shin
VLDB 2020

[41] MEGA: Multi-View Semi-Supervised Clustering of Hypergraphs
Joyce Jiyoung Whang, Rundong Du, Sangwon Jung, Geon Lee, Barry Drake, Qingqing Liu, Seonggoo Kang, and Haesun Park
VLDB 2020

[42] Hyperlink Classification via Structured Graph Embedding
Geon Lee, Seonggoo Kang, and Joyce Jiyoung Whang
SIGIR 2019 (Short Paper)

ACADEMIC SERVICE

Area Chair

- The Web Conference (WWW) 2027

Conference Reviewer

- ACM SIGKDD (KDD) 2023 – 2027
- AAAI 2024 – 2027
- ICLR 2025 – 2027
- AISTATS 2025 – 2027
- ACM CIKM 2022 – 2026
- Learning on Graphs (LoG) 2022 – 2026
- NeurIPS 2024 – 2026
- The Web Conference (WWW) 2024 – 2026
- ACM RecSys 2025 – 2026
- ICML 2025 – 2026
- ACM MM 2025 – 2026
- ACML 2025 – 2026
- ARR 2026
- SDM 2024

Journal Reviewer

- IEEE Transactions on Knowledge and Data Engineering (TKDE) 2023 – 2026
- Information Sciences 2024 – 2026
- IEEE Transactions on Network Science and Engineering (TNSE) 2024 – 2026
- ACM Transactions on Information Systems (TOIS) 2025 – 2026
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2025 – 2026
- Information Processing & Management 2025 – 2026
- Scientific Reports 2025 – 2026
- Machine Learning 2025 – 2026
- Neural Networks 2025 – 2026
- Patterns 2025 – 2026
- Transactions on Machine Learning Research (TMLR) 2026
- Journal of Machine Learning Research (JMLR) 2026
- Discover Artificial Intelligence 2026
- Expert Systems with Applications 2026
- Journal of Neuroscience Methods 2026
- Neurocomputing 2026
- Knowledge-Based Systems 2026
- Applied Soft Computing Journal 2026
- Information Fusion 2026
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS) 2023 – 2025
- The VLDB Journal 2023 – 2025
- PLOS ONE 2024 – 2025
- Data Mining and Knowledge Discovery 2024 – 2025
- Big Data Research 2024

Session Chair

- ACM CIKM 2024

AWARDS & HONORS

Gold Reviewer, ICML 2026 — May 2026
PhD Dissertation Award, KAIST College of Engineering — Feb. 2026
Best Paper Award, ICDM 2025 — Dec. 2025
One of the Best-Ranked Papers, ICDM 2025 — Dec. 2025
Best Survey Paper Award, PAKDD 2025 — Jun. 2025
Outstanding Reviewer, KDD 2025 — Dec. 2024
KAIST Outstanding Paper Award — Nov. 2024
Best Short Paper Candidate, RecSys 2024 — Oct. 2024
One of the Best-Ranked Papers, ICDM 2021 — Dec. 2021
Sungkyunkwan Presidential Award — Aug. 2019
Sungkyunkwan Software Scholarship (Full tuition, all semesters) — 2016 – 2019

TEACHING

Teaching Assistant

- KAIST AI506: Data Mining and Search — Spring 2021, 2023
- KAIST AI607: Graph Mining and Social Network Analysis — Fall 2020–2023
- KAIST AI617: Machine Learning for Robotics — Spring 2022
- SKKU CSE3036: Seminar in Computer Engineering — Fall 2019

Tutorial Presenter

- Mining of Real-World Hypergraphs: Patterns, Tools, and Generators — CIKM 2022, ICDM 2022, WWW 2023, KDD 2023
- Retrieval-Augmented Generation for LLM-Based Recommender Systems — KDD 2026, CIKM 2026

Guest Lecture

- Recent Advances in Recommender Systems (KAIST AI506) — May 2026

TALKS

- Understanding Complex Interaction Systems with AI: From Human Networks to AI Agent Societies — Yonsei University, Mar. 2026